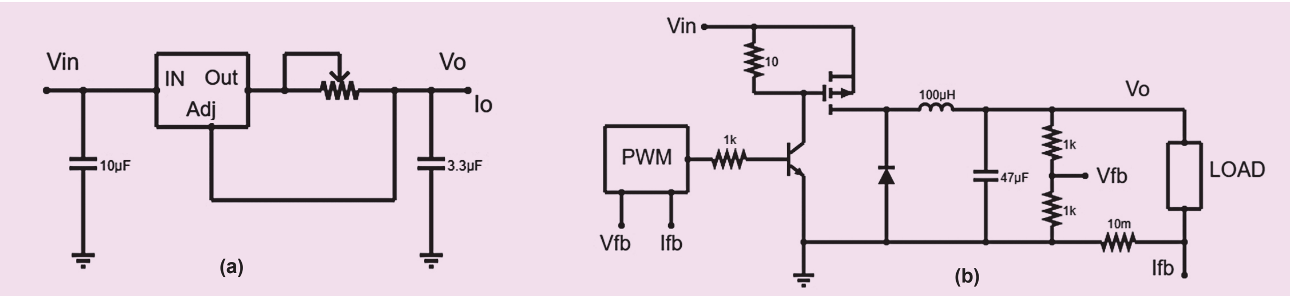

Fig. 2: (a) Adjustable voltage regulator IC used as constant current driver, (b) MOSFET switch based regulator for constant current control

load unless the impressed voltage is set to a value such that the required current flows against the load resistance. Thus, voltage control is an essential part of current control.

When the battery is in CC charging phase, this protection becomes part of a continuous control strategy, which aims at keeping the current value constant within the given battery terminal voltage envelope. This is generally achieved using a linear or switching regulator design, like the voltage control strategies shown above, where the quantity being controlled is current. While there are tens of ways to create a constant current source, only the simplest ones are presented below for reference.

In CV charging phase, the protection becomes a simpler limiting control problem, where a simple comparator circuit in hardware or comparison operators in the microcontroller code does the task. If the current flowing into the battery (or the load) increases beyond a pre-set limit, the designer can either choose to shut down the charging supply or reduce the impressed voltage to keep the current flowing within a limit.

Most switching power supplies provide overload protection (which is basically over-current protection), whereby the power supply shuts down the output for some time and then rechecks by starting the supply again. In the world of power supplies, this is called the hiccup mode.

In the worst case, the hiccup mode, with its repeated start-stop cycles, can cause permanent damage to some of the control electronic elements, such as op-amps and other control ICs. If the power is only being used for battery charging, then this might be alright. In other cases, where the power supply is shared by other sub-systems, it is best to use a linear current limiting provision of some kind to ensure longevity of the control electronics.

Charging/discharging over-temperature protection

This type of protection for batteries is generally part of the battery management systems. Batteries are electro-chemical products, and hence they are typically sensitive to temperature. In general, heightened temperatures for long times can cause permanent and fatal damage to their cells. This is true for all battery chemistries.

Li-ion chemistries are unique in the fact that they also have a lower temperature limit, below which their operation is not possible. So, for Li-ion batteries both low and high temperature protections are required. This protection is generally provided by using a surface temperature sensor, such as an RTD (resistance temperature device) that changes its resistance based on the surface temperature where it is attached. An electronic circuit is then used to convert this into a proportional voltage (for small distances) or current (for longer distances) that is fed to a comparison circuit or comparison operators in the microcontroller code, which then takes preventive action by disconnecting the charging or discharging paths.

During the charging process, over-temperature condition may occur if the charging current is much higher than recommended. For example, in a lead-acid battery, a 0.1 to 0.3°C charging rate is considered quite safe, while for a Li-ion battery, a 1°C rate is considered alright. Anything higher than these values may heat up the battery.

Over and above the basic preventive actions that the system should take, some systems may also provide active cooling or heating controls to ensure that the battery temperature stays within a prescribed range. Temperature is a relatively slowly changing quantity. In control systems parlance, temperature has a rather large time constant as compared to other physical quantities, such as voltage or current. So, temperature control in battery management systems is mostly implemented using simple on/off control with varying time intervals, depending on how close the temperature is to the limiting value.

Reverse polarity protection

Reverse polarity protection ensures that unintended high current does not flow into or out of the battery. During charging a battery may look like a load, and while discharging the battery acts as a source of energy. Connecting incorrect polarity of the battery to the charger results in a large potential difference and hence an almost uncontrolled current flow ensues, limited only by the